A POMDP Model of Rule Inference under Partial Observability for the Box task

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Rule Inference under Partial Observability

Box Task Setup. Subjects are given 5 boxes and 13 keys. The boxes are always lined up in the same order. The keys are given in an unordered pile. Keys have coloured key-tags. Each key tag has either a number or a shape. Boxes have colours. Each box also has at least one shape on it. Shapes are numbered from 1 to 5. Each face of a box, except for the front face, either has a shape or does not.

Misleading Instructions. Subjects are given instructions that to open all boxes they have to match boxes and key-tags by colour. A red key is demonstrated to open the red box. These instructions are incorrect and are introduced to induce a wrong prior belief. Unknown to the subjects, the true rule is to match the number of shapes on the box to the number on the key-tag.

Objective. Subjects are given the goal of opening all boxes within a limited time. We assume that people have a latent goal of inferring the rule governing box–key matching, but may also sample stochastically when uncertain.

Partial Observability. The setup is partially observable: subjects can only see part of each box. Initially, they observe the colour and shape of each box, but not the number of shapes. They know a lower bound on the number of shapes; the upper bound is always 5.

At each time t, subjects choose between:

- **Observe:** observe(i) reveals the exact number of shapes on box i
- **Attempt:** attempt(i, j) attempts to open box i with key j, where $i \in \{1, \ldots, 5\}$ and $j \in \{1, \ldots, 13\}$

Initial Box Configuration:

Box	Order	Colour	Shape	Lower Bound
B1	1	red	moon	1
B2	2	pink	cloud	1
B3	3	cream	diamond	4
B4	4	purple	heart	1
B5	5	teal	triangle	4

Keys:

Key	Colour	Shape	Number
red1	red		1
pink6	pink		6
grey2	grey		2
greycloud	grey	cloud	
orange4	orange		4
green3	green		3
bluestar	blue	star	
yellow5	yellow		5
greenheart	green	heart	
white7	white		7
triangleyellow	yellow	triangle	
diamondorange	orange	diamond	
purplearrow	purple	arrow	

True State (Unobserved).

Box	Order	Colour	Shape	Number
B1	1	red	moon	1
B2	2	pink	cloud	2
B3	3	cream	diamond	4
B4	4	purple	heart	3
B5	5	teal	triangle	5

Computational Model (POMDP)

We model this problem task as a hierarchical POMDP maintaining a belief $b(s)$ over environment states.

Each state s specifies the number of shapes on each box and which boxes are open:

$$b(s) = b(n_1, \ldots, n_5) \cdot \delta(o_1, \ldots, o_5),$$

where $n_i \in \{1, \ldots, 5\}$ and $o_i \in \{0, 1\}$.

The observable open states define the termination condition:

$$\sum_i o_i = 5.$$

Action Space

- **Observe box i:**

$$a = \text{observe}(i), \quad i \in \{1, \ldots, 5\}$$

- **Attempt (key, box) (i, j):**

$$a = \text{attempt}(i, j), \quad i \in \{1, \ldots, 5\}, \quad j \in \{1, \ldots, 13\}$$

Belief-Level Policy

$$\pi(a \mid b) \propto \exp\big(V(b, a)\big),$$

where $V(b, a)$ includes both inference and action value.

Value of Attempt Actions

We define:

$$V_{\text{attempt}}(a, b, h) \propto P(\text{success} \mid b, a, h) = \mathbb{E}_{s \sim b}\left[P(\text{success} \mid s, a, h)\right].$$

$P(\text{success} \mid s, a, h)$ is computed by the inference engine given state s, action a, and history h. This treats attempts as independently valuable contributions to task completion.

Value of Observe Actions

The value of observing box i is:

$$V_{\text{observe}}(b, \text{observe}(i)) = \mathbb{E}_{o \sim P(o|b,i)}\left[V(b')\right],$$

where b' is the updated belief.

We define:

$$V(b) = \max_a \mathbb{E}_{s \sim b}\left[P_{\text{engine}}(\text{success} \mid s, a, h)\right].$$

This corresponds to the value of the best attempt action under the current belief.

1 Assumptions

- The state space is enumerable given bounds:

$$5 \times 5 \times 2 \times 5 \times 2 = 500$$

  allowing exact belief updates (this may be expensive and approximate methods may still be useful).

- Following precedents from prior work, we use a single-step approximation for $V(b)$ rather than full multi-step planning.

- The inference engine has parameters θ and depends on history h. The observation process has no free parameters.

- We assume uniform action costs and omit them. This may bias the model toward excessive observation. Intuition: If subjects do not immediately observe boxes, this may be explained by strong prior beliefs induced by misleading instructions (e.g., colour matching). In that case, observation penalties are unnecessary. If behavior still shows under-observation even after priors are unlearned, this suggests an additional cost (e.g., social or time cost) associated with observation.

- TBD: empirically estimate the ratio of Observe vs Attempt actions across participants.